Environmental Sciences_2024_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. What are causes of acid rain and what is its effects on natural environment?
2. What do you understand by GCM?
3. What are In-situ and Ex-situ conservation method of biodiversity?
4. Graphically explain the relationship between BOD and DO.
5. What are the various sources of water pollution?
6. Mention name of few gases generated from Incineration plant.
7. Mention various types of noise pollution and its effects on human being.
8. What authorizes the Central Government of India in Environmental Protection Act, 1986?
9. What are the various types of earthquake waves generated in the earth surface?
10. Mention the types and function of various components of ecosystem.
11. Mention the various elements of air pollution as per NAAQS.
12. What is meant by 5-day BOD and Ultimate BOD?

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Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

2. (i) Explain with neat diagram the internal structure of the Earth.
(ii) What do you mean by rock cycle? [5]
3. (i) What do you mean by maximum mixing depth?
(ii) Differentiate between Ambient lapse rate and Adiabatic lapse rate. [5]
4. (i) Define flood and how can it be mitigated and controlled?
(ii) What do you understand by anthropogenic environmental degradation? [5]
5. (i) Write the equation for population growth and explain the notations.
(ii) How to calculate 'Natural Increase Rate'? [5]
6. (i) How arsenic get dissolved in groundwater in the aquifer system? [5]
(ii) How many districts in West Bengal are exposed to groundwater arsenic contamination?

Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

7. (i) What are different types of solid wastes?

(ii) What are the methods of solid waste disposal and management?

(iii) What are the advantages and disadvantages of Land filling of solid waste? [4+6+5]

8. (i) What are Nutrient cycles and how are they important for our environment?

(ii) Draw a flowchart for Nitrogen Cycle.

(iii) Explain Biogeochemical cycle by a diagram. [4+6+5]

9. (i) Mention various types of greenhouse gases and its effects on terrestrial ecosystems.

(ii) How global warming and climate change are related?

(iii) What is short wave and long wave radiation? [6+5+4]

10. (i) Why sustainable growth is considered modern concept of development?

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(ii) How population growth in India can be controlled?

(iii) If the present population of the World is 8.00 billion, what would be the population of the World after 25 years with a growth rate of 1% per year? [5+3+7]

11. (i) What is a groundwater aquifer, aquiclude and aquitard. Explain with neat diagram.

(ii) How groundwater is recharged?

(iii) What do you understand by Rooftop rainwater harvesting? [6+4+5]